# Hybrid Multistrategy Remora Optimization Algorithm-Based Band Selection for Hyperspectral Image Classification

Heming Jia and Zhengbang Li

***Abstract*— Hyperspectral image (HSI) is celebrated for its detailed spectral information but faces significant challenges in dimensionality reduction stemming from excessive spectral dimensions. Band selection (BS) is a critical technique in dimension reduction, aiming to identify an optimal subset of spectral bands with minimal redundancy and maximal feature separability. Swarm intelligence methods are renowned for their flexibility and high performance in optimization problems. These methods have been extensively introduced by scholars to address BS tasks in hyperspectral imaging. Among these, the remora optimization algorithm (ROA) stands out for its exceptional optimization proficiency. However, its conventional evolutionary operators are susceptible to local optimum stagnation. Therefore, a novel BS method based on an improved ROA, termed IROA-BS, is proposed for HSI classification. First, an evaluation function is designed to estimate the class separability and redundancy of selected band subsets. Second, the hybrid evolutionary operators are intended to diversify potential solutions, while a multistage mutation module is implemented to circumvent local optimum stagnation. Furthermore, a guided learning strategy is utilized to fine-tune the equilibrium between exploration and exploitation processes. The effectiveness of the proposed IROA-BS method is rigorously validated across three widely recognized hyperspectral remote sensing image datasets. Comparative analysis with the other advanced BS methods and swarm intelligence techniques validates the superiority and efficacy of the IROA-BS method in HSI BS applications.**

***Index Terms*— Band selection (BS), guided learning strategy (GLS), hyperspectral image (HSI), multistage mutation, remora optimization algorithm (ROA).**

## I. Introduction

HYPERSPECTRAL remote sensing represents a cutting-edge domain within the sphere of imaging spectroscopy. Hyperspectral images (HSIs) are acquired using hyperspectral sensors that capture images across narrow spectral bands spanning a broad frequency spectrum, offering a richness of detail surpassing that of conventional imagery [1]. This capability to discern objects with minute or closely aligned spectral differences renders HSIs immensely advantageous across diverse specialties, such as biomedicine [2], agriculture [3], and geology [4]. Nonetheless, the inherent high-dimensional characteristics of HSIs entail considerable computational burdens [5]. Furthermore, the proliferation of spectral channels tends to engender excessive redundant data and noise that potentially degrades performance in various applications, including classification [6], anomaly detection [7], and change detection [8]. Consequently, investigating and devising more efficacious dimensionality reduction (DR) techniques remains a matter of paramount importance.

Within the ambit of contemporary research, DR methodologies are typically bifurcated into two principal categories: feature selection [9] and feature extraction [10]. Feature selection, recognized in the context of hyperspectral imaging as band selection (BS), entails the identification and consolidation of the most informative spectral bands from the HSI dataset, thus formulating a condensed band subset [11]. Conversely, feature extraction endeavors to compress the high-dimensional space of an HSI into a lower-dimensional representation by synthesizing a novel constellation of attributes [12]. Distinct from feature selection, feature extraction modifies the original attributes of the dataset, potentially leading to the forfeiture of vital information [13]. Consequently, BS methods are prevalently applied to address DR challenges in hyperspectral remote sensing imagery. Current research in hyperspectral BS predominantly harnesses swarm intelligence techniques, exemplified by the employment of an improved genetic algorithm [spectral-spatial genetic algorithm (SSGA)] [14] integrated with superpixel technology for BS; the proposition of a hybrid gray wolf optimizer (HGWO) [15] incorporating an adaptive convergence factor for BS tasks; the development of an adaptive multistrategy particle swarm optimization (AMSPSO) [16] that leverages linear discriminant value and mean mutual information for the election of an efficacious HSI band subset; and the deployment of a heterogeneous cuckoo search-based (HCS) approach [17] paired with a matched filter for HSI classification.

In the contemporary field of research, swarm intelligence algorithms, also distinguished as metaheuristic algorithms, are lauded for their adaptability and efficacy in problem-solving, marking a significant advancement within the domain of optimization challenges [18]. These methods principally undergo

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Heming Jia is with the School of Information Engineering, Sanming University, Sanming 365000, China (e-mail: jiaheming@fjsmu.edu.cn).

Zhengbang Li is with the School of Computer Science and Mathematics, Fujian University of Technology, Fuzhou 350118, China (e-mail: 2221308067@smail.fjut.edu.cn).

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two pivotal stages [19]: exploration and exploitation. The exploration phase is dedicated to the identification of candidate solutions within the global search space, whereas exploitation targets the refinement of solutions in the vicinity of the currently identified optimum. Rooted in the principles of natural phenomena and biological intelligence, these algorithms are categorized into four distinct groups [20]: evolution-based algorithms (EAs), physics-base algorithms (PAs), human-based algorithms (HAs), and swarm-based algorithms (SAs).

EA is fundamentally influenced by the dynamics of natural evolution. During each phase, the genetic algorithm (GA) [21] employs genetic operations, including mutation, crossover, and selection, to cultivate novel candidate solutions. Noteworthy algorithms within this category encompass evolutionary programming (EP) [22] and differential evolution (DE) [23]. PA derives methodologies from the simulation of physical phenomena, like gravitational forces and energy dynamics, to ascertain optimal solutions [24]. An illustration of this is the gravitational search algorithm (GSA) [25], which is predicated on the principle of gravity, where potent forces are exerted among proximal mass populations in contrast to the lesser forces between more distant ones. HA adopts the intricacies of human social phenomena to address complex optimization challenges [26]. An innovative method proposed in this spectrum is the growth optimizer (GO) [27], which mirrors the human processes of learning and reflection. SA draws inspiration from the organized and collective behaviors observed in social entities such as ants, bees, birds, and fish [28]. A cornerstone algorithm in this domain, particle swarm optimization (PSO) [29] takes its cue from the flocking behavior of birds. Meanwhile, beluga whale optimization (BWO) [30] introduces a mechanism reminiscent of whale falls, aimed at purging suboptimal solutions. In addition, the coati optimization algorithm (COA) [31] simulates the foraging strategies of coatis and their evasion from predators.

In this article, the remora optimization algorithm (ROA) [32] is strategically utilized to address the BS conundrum. ROA is a relatively novel addition to the arsenal of swarm intelligence techniques, gaining recognition for its proficiency in navigating through complex optimization landscapes and engineering quandaries [33], [34]. Nonetheless, a notable limitation is observed in the evolutionary operator's search capability within the foundational ROA framework. Moreover, the no-free-lunch (NFL) theorem [35] postulates the impossibility of a singular algorithm's aptitude to universally resolve all categories of problems. Consequently, this article introduces a novel BS method based on improved ROA (IROA-BS), designed with hybrid evolution operators, multistage mutation, and a guided learning approach, tailored to effectively tackle the HSI classification.

The main contribution of this article can be summarized as follows:

To assess the quality of the HSI band subset solution in every generation within the presented IROA-BS methodology, a dedicated evaluation function has been devised. This function is responsible for determining the class separability and redundancy of the candidate band subset. BS methods that incorporate swarm intelligence algorithms and utilize the proposed evaluation function exhibit notably enhanced performance relative to other BS techniques.

A hybrid evolution module has been designed to augment the population's diversity and enhance searchability performance. Through the incorporation of diverse evolution operators, the population is empowered to unearth more satisfactory solutions. Consequently, this methodology facilitates the identification of more efficacious HSI band subsets, featuring diverse combinations.

A multistage mutation module, incorporating three distinct mutation operators, is proposed to fortify the system's ability to sidestep local entrapment. When local optima stagnation occurs, the efficiency of evolution operators tends to diminish throughout the iterative process. To mitigate this issue, three tailored mutation operators are integrated at various stages of evolution.

A novel mechanism, the guided learning strategy (GLS), tailored for swarm intelligence algorithms, is employed to elevate the update performance of the proposed methodology. This strategy is pivotal in achieving an optimal balance between exploration and exploitation, crucial for enhancing performance. Consequently, GLS leverages historical learning experiences to finely tune this balance.

The remainder of this article is structured as follows: Section II outlines the foundational principles of the ROA method. Section III provides an in-depth examination of the proposed IROA-BS method. Detailed accounts of the experimental setup and comparative analyses are presented in Section IV. Section V delves into the performance analysis of the proposed BS framework from various perspectives. The article culminates in Section VI with a conclusion.

## II. Basic ROA Method

Remoras, belonging to the family of ray-finned fish, have developed a unique oval, sucker-like organ enabling them to create suction. This evolutionary adaptation allows remoras to securely attach to the skin of larger marine animals, facilitating their transportation across various marine regions for foraging, by leveraging the locomotion of their hosts. In ROA, the behavioral patterns of humpback whales and sailfish are strategically utilized as exemplars for the remoras. The dynamic locomotion of these marine hosts is meticulously emulated, thereby facilitating the simulation of the remoras' parasitic movement. This approach allows for the replication of the remoras' adept ability to leverage the mobility of larger marine species for their navigational purposes. Moreover, most swarm intelligence algorithms consider the optimal solution as food, prey, or leader of the population to iteratively evolve the current population. Additionally, akin to most swarm intelligence algorithms, the optimal solution $p_{\text{best}}$ within the ROA framework is conceptualized as food, prey, or the leader of the population, guiding the iterative evolution of the current population.

Based on the whale optimization algorithm (WOA) [36], the whale exhibits a distinctive foraging behavior dubbed the bubble-net attacking strategy, which enables it to amass a significant quantity of prey. In a symbiotic relationship, the remora, or parasitic fish, leverages this method to its

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advantage. A portion of the remora population utilizes the whale as a host for predation, as shown in the following:

$$p_i^{t+1} = \varepsilon \cdot \left| p_{\text{best}} - p_i^t \right| + p_{\text{best}} \tag{1}$$

$$\varepsilon = \cos\left(2\pi \cdot \frac{t}{T}\right) \exp\left[1 - r \cdot \left(2 + \frac{t}{T}\right)\right] \tag{2}$$

where $p_i^{t+1}$ represents the solution of the next generation, $\varepsilon$ is a random parameter that is generated by the spiral function, $r$ indicates a random number in the range (0, 1), $t$ represents the current iteration, and $T$ is the maximum number of iterations.

Leveraging the sailfish population's foraging patterns, a segment of the remora population has adapted to search for sustenance over an extended area. Based on the sailfish optimizer (SFO) [37], the parasitic model is conceptualized as detailed below

$$p_i^{t+1} = p_{\text{best}} - \left[\frac{r}{2}\left(p_{\text{best}} - p_{\text{rand}}^t\right) - p_{\text{rand}}^t\right] \tag{3}$$

where $p_{\text{rand}}^t$ represents a randomly selected individual in the $t$th generation.

Beyond the realm of parasitic predation, models have been constructed to represent independent foraging behaviors. These are encapsulated by two foundational evolutionary operations: the experience attack and the host feeding strategy. The former strategy emulates the remora's autonomous search, guided by the experiences of the preceding generation. The mathematic modal is depicted as follows:

$$p_i^{t+1} = p_i^t + r_n \cdot \left(p_i^t - p_i^{t-1}\right) \tag{4}$$

where $p_i^{t+1}$ represents the history experience of the $i$th remora, $r_n$ is a random number that obeys normal distributions.

The second strategy models the behavior of remoras foraging in proximity to their host. The model of host feeding is characterized as outlined below

$$p_i^{t+1} = p_i^t + \eta\left(p_i^t - C \cdot p_{\text{best}}\right) \tag{5}$$

$$\eta = 2(2r - 1)\left(1 - t/T\right) \tag{6}$$

where $\eta$ represents the random parameter, which decreases linearly and controls the evolutionary size of the remora in the host feeding stage, and $C$ indicates the volume of the host.

## III. Proposed Method

### A. Evaluation Function

Centering on the optimization protocols of the swarm intelligence algorithm within HSI BS, it is imperative to devise an evaluation function. This function assesses the image quality of the chosen data subset to ultimately enhance classification efficacy. Accordingly, an evaluative model is conceived to quantify the correlation and superfluous information inherent in the acquired subsets.

In this article, we employ an efficient evaluation criterion, denoted as $f_1$ [38] which is predicated on the distinct class separability and the compactness within each class, to assess correlation. The $f_1$ criterion amalgamates the correlation between interclass and intraclass dispersions aiming to attain optimal class separability. Pursuant to the aforementioned objective, it is essential to maximize the interclass scatter matrix $S_b$ and minimize the intraclass scatter matrix $S_w$. Consequently, the correlation evaluation criterion $f_1$ is constructed as follows:

$$f_1 = \text{tr}\left(S_w^{-1} S_b\right) \tag{7}$$

$$S_b = \sum_{k=1}^{K} p_k \cdot (\mu_k - \bar{\mu})\ (\mu_k - \bar{\mu})^T \tag{8}$$

$$S_w = \sum_{k=1}^{K} p_k \cdot cov\ (x_k) \tag{9}$$

where $p_k$ represents the prior probability of the $k$th category, $\mu_k$ is the mean spectrum vector of the $k$th category, $\bar{\mu}$ denotes the mean of $\mu_k$, $\text{cov}(x_k)$ indicates the covariance matrix of the $k$th category, $x_k$ is the spectrum matrix of the $k$th category, and tr denotes the trace of the matrix.

The second evaluation criterion $f_2$ [39] is predicated upon information entropy, a concept frequently employed in BS for estimating redundancy. Typically, a lower degree of redundant information necessitates a higher entropy value. The particulars of $f_2$ are specified as follows:

$$f_2 = -\sum_{x_j \in X} p(x_j) \log p(x_j) \tag{10}$$

$$p(x_j) = \frac{\text{hist}(x_j)}{h \times w} \tag{11}$$

where $X$ is a set of all spectrum bands, $x_j$ denotes the selected band, $p(x_j)$ represents the probability density in the $j$th band image, $\text{hist}(x_j)$ indicates the grayscale value in the $j$th band image, and the $h \times w$ is the size of the band image.

Furthermore, to diminish the number of selected bands, a band number control function, denoted as $f_3$, is implemented. This function is defined as follows:

$$f_3 = |N - \varphi \cdot n_s| \tag{12}$$

where $N$ represents the total band number of HSI, $n_s$ denotes the band number in the selected subset, and $\phi$ is a band number adjust parameter.

As a result, the ultimate evaluative function model for BS within the swarm intelligence methodology, as applicable to solution $p_i$ is designed as shown in the following:

$$\min f(p_i) = -\frac{f_1 + f_2 - n_s f_3}{n_s}. \tag{13}$$

### B. Proposed IROA-BS Method

In this section, we delve into the intricacies of the proposed IROA-BS method, including its procedural nuances. The method, distinguished by three primary enhancements, is expounded upon in Sections III-B2–III-B4. The structural composition of the IROA-BS methodology is illustrated in Fig. 1, while the corresponding pseudocode is shown in Algorithm 1.

*1) Initialization:* Presuming that $X$ represents the HSI dataset, the $x$ refers to each individual band, which can be described as $X = \{x_1, x_2, \ldots, x_n\}$, where $n$ indicates the total count of bands in the HSI. During the initial phase, a series

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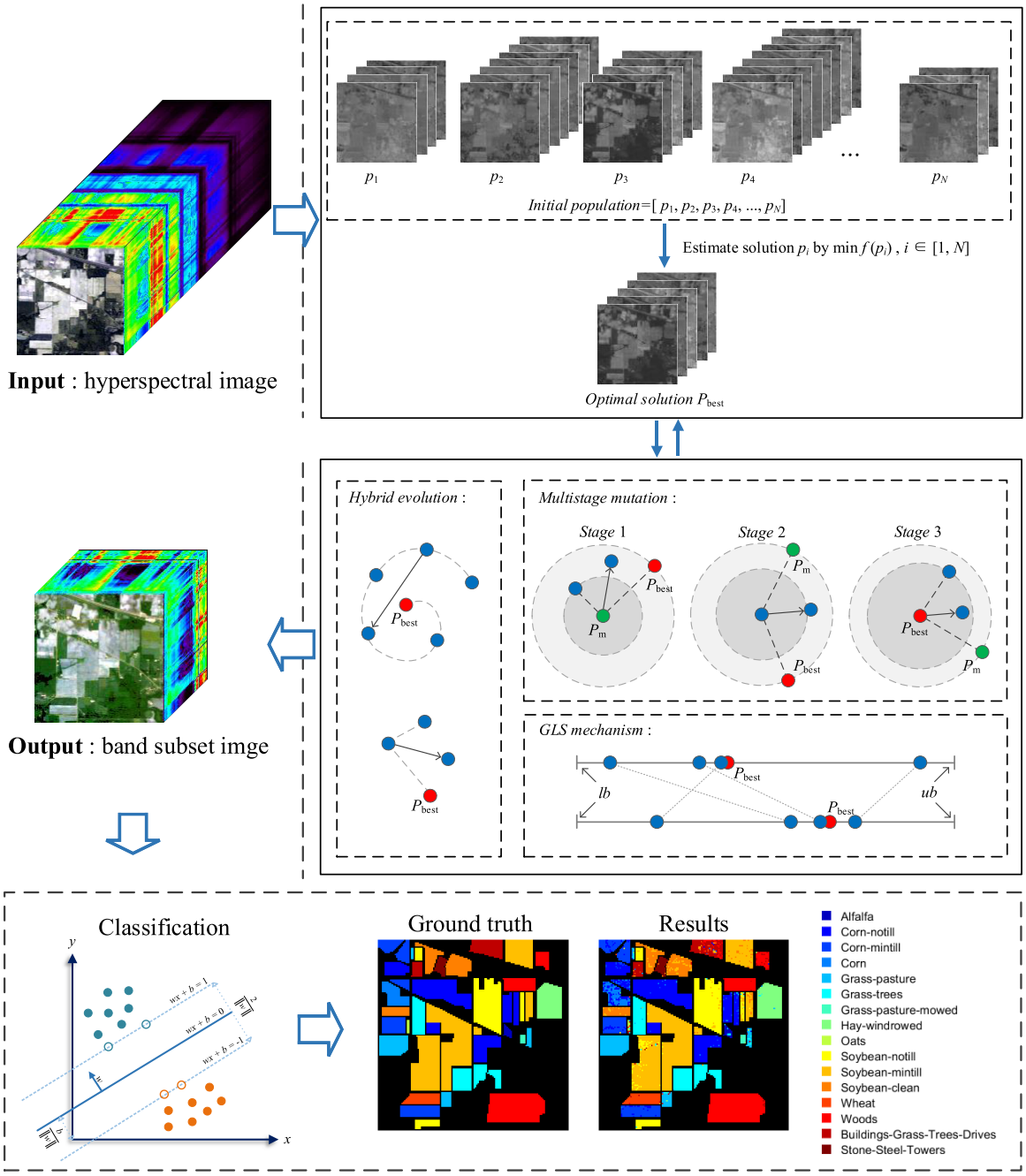



Fig. 1. Framework of the proposed IROA-BS method.

of binary codes are generated in a stochastic manner, each indicating the selection status of a band. For instance, one potential initial solution when dealing with an HSI comprising 100 bands might be represented as "110, . . . , 000, . . . , 001," which implies that the subset of bands selected consists of $\{x_1, x_2, x_{100}\}$. Fig. 2 shows the configuration of the solution and exhibits two distinctive solution samples.

*2) Hybrid Evolution Module:* Integrating diverse optimization algorithms to bolster convergence efficiency is a prevalent strategy in the advancement of swarm intelligence algorithms [40], [41]. The honey badger algorithm (HBA) [42], an innovatively proposed technique embodying the foraging conduct of honey badgers, has demonstrated superior optimization capabilities in addressing high-dimensional

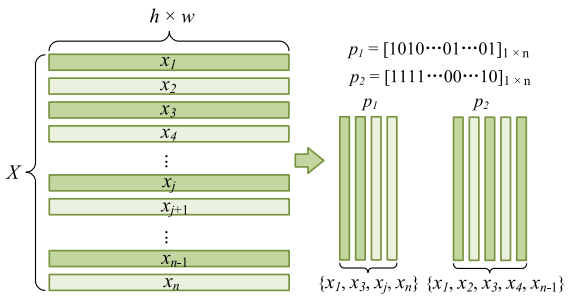



Fig. 2. Structure of the population and two samples of the selected band subset.

problems. Consequently, this study employs two principal evolutionary operators, specifically during the "digging

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**Algorithm 1** Pseudocode of IROA-BS

**Input**:
$X$: hyperspectral image data;
$N$: size of population;
$T$: maximum number of iterations;
$MR$: mutation rate;
$Gmax$: size of learning experience;
$\alpha$: thresdhold of feedback result.
**Output**:
$p_{best}$ : the optimal solution;

```
Initialize the population P_i(i = 1, 2, ..., N);
Evaluate the population by f and update p_best;
while t < T do
    for i = 1 to N do
        Update p_i by hybrid evolution operators (1), (3), (14), and (18);
        if r < MR then
            Calculate mutate p_i according to (19);
        end
        if r < 0.5 then
            Update p_i by experience attack modal (4);
        else
            Update p_i by host feeding modal (5);
        end
    end
    t = t + 1;
    Evaluate the population by f and update p_best;
    Store p_i into p_st as the learning experience;
    for g = 1 to Gmax do
        Update feedback result FR by (22);
        Update p_i according to guide learning strategy by (21);
    end
    Evaluate the population by f and update p_best;
end
return the optimal solution p_best.
```

phase" and "honey phase" of the HBA, to update the population.

The first operator employs a cardioid curve to replicate the digging activity of a honey badger, with $\beta$ denoting the cardioid factor. Furthermore, an intensity factor $I$ is designed to articulate the level of focus between the individual and the object of optimal search. Based on the above description, the operator is defined as follows:

$$p_i^{t+1} = p_{\text{best}} + 6F \cdot I_i \cdot p_{\text{best}} + F \cdot \delta \cdot \left(p_{\text{best}} - p_i^t\right)\beta \quad (14)$$

$$\beta = |\cos(2\pi r_1)[1 - \cos(2\pi r_2)]| \quad (15)$$

$$I_i = r_1 \cdot \left(p_i^t - p_{i+1}^t\right)^2 \Big/ 4\pi \cdot \left(p_{\text{best}} - p_i^t\right)^2 \quad (16)$$

$$\delta = 2r_1 \cdot \exp\left(-t/T\right) \quad (17)$$

where $F$ indicates a direction flag, which is stochastically generated to 1 or $-1$, $\delta$ is a nonlinear decreasing parameter, and $r_1$ and $r_2$ are random fractions in the range (0, 1).

The second operator mimics the guiding behavior of a honey badger, transitioning from the current solution toward the optimal solution., which is defined as shown in the following:

$$p_i^{t+1} = p_{\text{best}} + F \cdot \delta \cdot \left(p_{\text{best}} - p_i^t\right). \quad (18)$$

To augment the diversity of the population, the hybrid evolution module, as defined by (1), (3), (14), and (18), amalgamates four evolutionary operators. Each operator is randomly chosen to compute the progeny solution within a single generation. The hybrid evolution module is incorporated across every iteration for enriched population variety.

*3) Multistage Mutation Module:* During the initial stages of evolution, it is often the case that evolutionary operators inadvertently steer the population toward a local entrapment scenario. Consequently, a multistage mutation module has been conceptualized with the goal of mitigating the incidences of being ensnared in local optima. This module comprises three mutation operators, each tailored to the evolutionary stage-early, middle, and late. The design of the module is defined as follows:

$$p_i^{t+1} = \begin{cases} p_m + \theta\left(p_i^t - p_m\right) + \hat{\theta}(p_{\text{best}} - p_m), & t \in \left(0, \frac{T}{3}\right] \\ p_i^t + \theta\left(p_m - p_i^t\right) + \hat{\theta}\left(p_{\text{best}} - p_i^t\right), & t \in \left(\frac{T}{3}, \frac{2T}{3}\right] \\ p_{\text{best}} + \theta\left(p_i^t - p_{\text{best}}\right) + \hat{\theta}(p_m - p_{\text{best}}), & t \in \left(\frac{2T}{3}, T\right] \end{cases} \quad (19)$$

$$\theta = \exp\left(-t/T\right), \quad \hat{\theta} = 1 - \theta \quad (20)$$

where $p_m$ represents a mutation solution, which is calculated by the object boundary, and $\theta$ is a nonlinear decreasing parameter, which can adjust the mutation distance of the operators. Moreover, the mutation rate $MR$ is set to 0.05 in this work.

*4) Guide Learning Strategy:* Achieving a balance between exploration and exploitation constitutes a fundamental aspect in the development of an efficacious metaheuristic algorithm. On this account, this study harnesses a GLS [43] to enhance the optimization prowess of ROA. GLS is a novel proposed mechanism for updating populations, draws inspiration from the educational process of knowledge acquisition. In [43], GLS employs historical solutions $p_{\text{st}}$ as learning experiences, leveraging these to calculate the feedback result FR which serves to navigate the equilibrium between the stages of exploration and exploitation. Moreover, Gmax represents a hyperparameter that constrains the quantity of $p_{\text{st}}$, while $\alpha$ denotes a threshold value for FR. The mathematical model of GLS is defined as follows:

$$p_i^{t+1} = \begin{cases} p_{\text{best}} + \tan(r \cdot \pi)(\text{ub} - \text{lb})\text{FR}^{-1}, & \text{FR} > \alpha \\ \text{lb} + r \cdot (\text{ub} - \text{lb}), & \text{FR} \le \alpha \end{cases} \quad (21)$$

$$\text{FR} = \lambda \cdot \text{std}(p_{\text{st}})(\text{ub} - \text{lb})^{-1} \quad (22)$$

where $r$ represents a random number within the range (0, 1), ub and lb signify the upper and lower bounds of the solution, respectively. $\lambda$ is a constant value for normalizing FR, and std denotes the function used for computing the standard deviation. Additionally, Gmax and $\alpha$ are established values set to 5 and 10, respectively.

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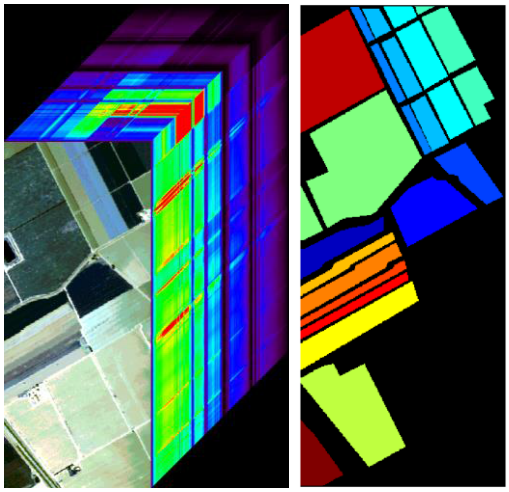


Fig. 3. HSI cube and ground truth of Salinas. (a) HSI cube. (b) Ground truth.

*5) Binary Mapping Function:* To acquire binary solutions for the BS problem, IROA-BS employs a sigmoid function to map the candidate solutions. The mapping modal is defined as follows:

$$\mathrm{bp}_{i,j} = \begin{cases} 0, & \mathrm{sigmoid}\left(p_{i,j}\right) > r \\ 1, & \text{otherwise.} \end{cases} \tag{23}$$

where $\mathrm{bp}_{i,j}$ represents band $j$th in $i$th binary solution and $r$ is a random number in the range (0, 1).

## IV. Experiments and Results

In this article, three well-established HSI datasets Salinas, Pavia Center, and Pavia University are utilized to ascertain the efficacy of band subset selection by the proposed IROA-BS method compared to other BS techniques.

### A. Description of Datasets

*1) Salinas Dataset:* Salinas is a well-known HSI dataset. This dataset was acquired over the Salinas Valley in California employing the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. It encompasses 224 spectral reflectance bands, covering 16 distinct classes within a wavelength spectrum ranging from 400 to 2500 nm. Each band is represented with a resolution of 512 × 217 pixels. Importantly, 20 water absorption bands, specifically bands 108–112, 154–167, and band 224, are excluded from the analysis. Consequently, 204 spectral bands are utilized for experimental purposes. Fig. 3 shows the image cube and ground truth of Salinas.

*2) Pavia Center Dataset:* The Reflective Optics System Imaging Spectrometer (ROSIS) sensor was used to gather this dataset in the urban area of Pavia, Italy. Comprising nine distinct classes, the original collection incorporates 115 bands. Upon the removal of 13 bands identified as noise, the final dataset consists of 102 spectral bands. The dimensions of each band's image are 1096 × 715 pixels, with a geometric resolution of 1.3 m. The spectral wavelength of the bands extends from 430 to 860 nm. Fig. 4 shows the image cube and ground truth of Pavia Center.

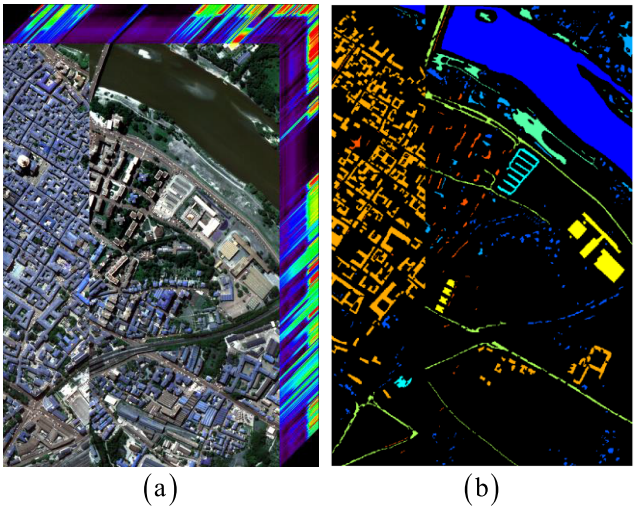


Fig. 4. HSI cube and ground truth of Pavia Center. (a) HSI cube. (b) Ground truth.

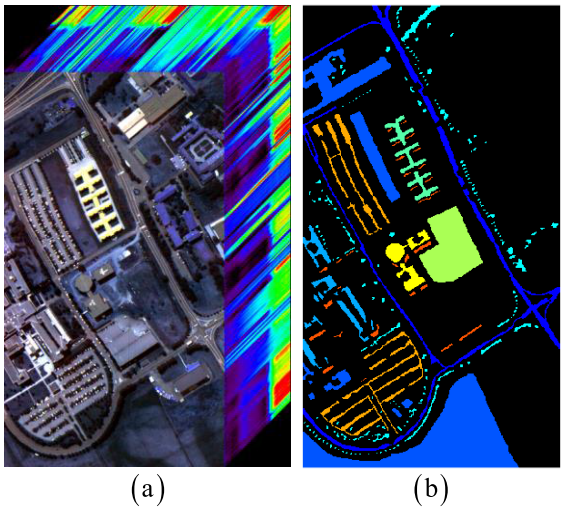


Fig. 5. HSI cube and ground truth of Pavia University. (a) HSI cube. (b) Ground truth.

*3) Pavia University Dataset:* Pavia University was acquired using the ROSIS sensor in the vicinity of the University of Pavia, Italy. It delineates nine land cover classes of interest, with 103 bands retained for experimental analysis. The dimensions of each band's image are 610 × 340 pixels. Fig. 5 shows the image cube and ground truth of Pavia University.

### B. Experiment Description

To assess the effectiveness of the proposed IROA-BS method, a range of BS techniques were employed to execute a comparative analysis. The assessment incorporated three conventional BS approaches: a classic ranking-based method maximum variance principal component analysis (MVPCA) [44], a fast density-peak clustering method enhanced fast density-peak-based clustering (E-FDPC) [45], and a multicriteria-based method subspace partition multicriteria-based band selection (SP-MCBS) [46]. In addition, four swarm intelligence optimization approaches were incorporated to perform BS tasks, including PSO [29], DE [23], sine

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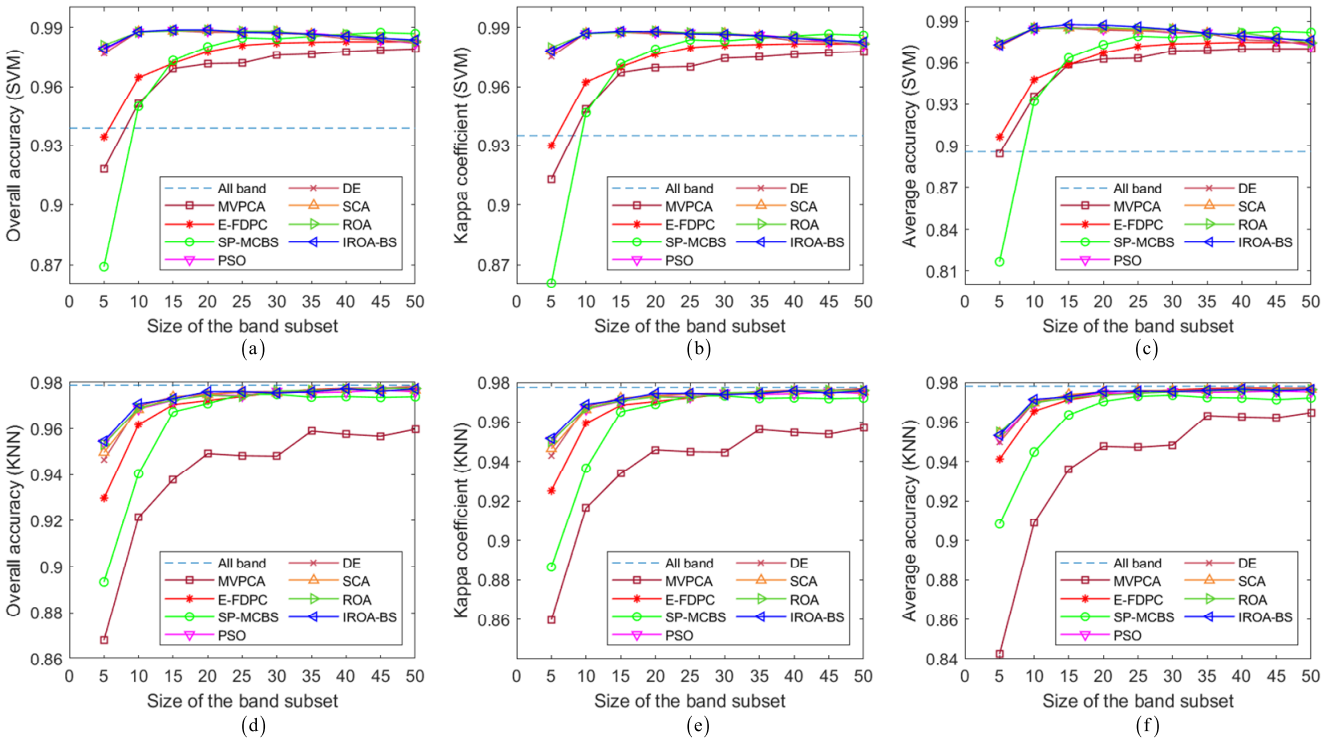



Fig. 6. Classification curves of different number of the selected bands on Salinas. (a) OA curves by SVM. (b) Kappa curves by SVM. (c) AA curves by SVM. (d) OA curves by KNN. (e) Kappa curves by KNN. (f) AA curves by KNN.

TABLE I
PARAMETERS OF THE INTELLIGENCE OPTIMIZATION METHODS

| Method | Parameter | Value |
|---|---|---|
| PSO | *w, C1, C2* | [0.2, 0.9], 1.5, 1.5 |
| DE | *F, CR* | 0.25, 0.8 |
| SCA | *a* | 2 |
| ROA | *C* | 0.5 |
| IROA-BS | *MR, Gmax, a* | 0.05, 5, 10 |

cosine algorithm (SCA) [47], and the traditional ROA [32]. In addition, a condition with all bands is employed to conduct the comparative trials. The settings for population size and maximum iterations were configured to 30 and 100, respectively. Parameters pertinent to these swarm intelligence algorithms are detailed in Table I. Furthermore, the number of evaluations was modulated under the iterations.

Following the acquisition of band subsets, two classifiers, support vector machine (SVM) and k-nearest neighbor (KNN), were deployed for training and predictive analysis. Each methodology was executed independently ten times. The training dataset was randomly constituted from 10% of the overall dataset. The SVM employed a Gaussian kernel function, with both the penalty coefficient and kernel parameter set to 1. Concurrently, the $k$-value for KNN was established at 8. Moreover, the predictive outcomes were evaluated utilizing metrics such as Overall Accuracy (OA), Kappa Coefficient (Kappa), and Average Accuracy (AA). All empirical investigations were executed on a PC equipped with an Intel Core i7-11700 CPU processor and 16 GB of RAM, utilizing MATLAB 2021a.

### C. Experiment Results

*1) Salinas:* The verification of the performance across various sized band subsets presents a challenge. Consequently, the number of selected bands within these subsets has been meticulously refined to range from 5 to 50 on the Salinas dataset. This refinement facilitates a direct comparison with an identical number of selected bands. The classification outcomes of the proposed IROA-BS method, along with various other methods as applied to the Salinas dataset, are depicted in Fig. 6. It is observed that, particularly when the size of the band subset is minimal, the IROA-BS method along with other swarm intelligence techniques demonstrates superior performance in BS compared to the MVPCA, E-FDPC, and SP-MCBS methods.

In the case of the SVM classifier, the IROA-BS and comparable methods exhibit the "Hughes" phenomenon wherein accuracy diminishes with an increase in the number of bands, particularly when their size exceeds 15. It is noted that when the size of the band subset exceeds 40, the performance of IROA-BS is relatively weaker. Nevertheless, in most scenarios, the results continue to outperform those attained by ranking-based methodologies. Notably, IROA-BS exhibits optimal performance when the size ranges between 15 and 30, as illustrated in Fig. 6(c). In the context of the KNN classifier, the performance of the BS methods stabilizes as the subset size increases. The comparative analysis indicates that IROA-BS maintains a marginal yet superior advantage over other methodologies. Meanwhile, MVPCA demonstrates comparatively reduced accuracy.

The details on the 20 selected bands are delineated in Table II. According to the findings, the IROA-BS method

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TABLE II
CLASSIFICATION ACCURACY OF PROPOSED IROA-BS AND DIFFERENT METHODS ON SALINAS

| Method | SVM | | | KNN | | |
|---|---|---|---|---|---|---|
| | OA (%) | Kappa (%) | AA (%) | OA (%) | Kappa (%) | AA (%) |
| All band | 93.90±0.12 | 93.52±0.13 | 89.56±0.24 | 97.88±0.12 | 97.75±0.13 | 97.80±0.13 |
| MVPCA | 97.16±0.09 | 96.97±0.09 | 96.28±0.14 | 94.92±0.13 | 94.60±0.14 | 94.77±0.21 |
| E-FDPC | 97.77±0.22 | 97.63±0.24 | 96.75±0.36 | 97.22±0.12 | 97.05±0.12 | 97.38±0.14 |
| SP-MCBS | 98.02±0.21 | 97.90±0.22 | 97.33±0.24 | 97.09±0.07 | 96.91±0.08 | 97.05±0.14 |
| PSO | 98.82±0.21 | 98.75±0.22 | 98.45±0.31 | 97.48±0.23 | 97.31±0.25 | 97.47±0.19 |
| DE | 98.76±0.14 | 98.68±0.15 | 98.38±0.26 | 97.50±0.35 | 97.34±0.37 | 97.52±0.24 |
| SCA | 98.85±0.19 | 98.78±0.20 | 98.50±0.33 | 97.48±0.32 | 97.32±0.34 | 97.53±0.24 |
| ROA | 98.86±0.10 | 98.79±0.10 | 98.56±0.15 | 97.45±0.19 | 97.28±0.20 | 97.36±0.19 |
| IROA-BS | **98.88±0.19** | **98.81±0.20** | **98.73±0.33** | **97.60±0.29** | **97.45±0.30** | **97.55±0.19** |

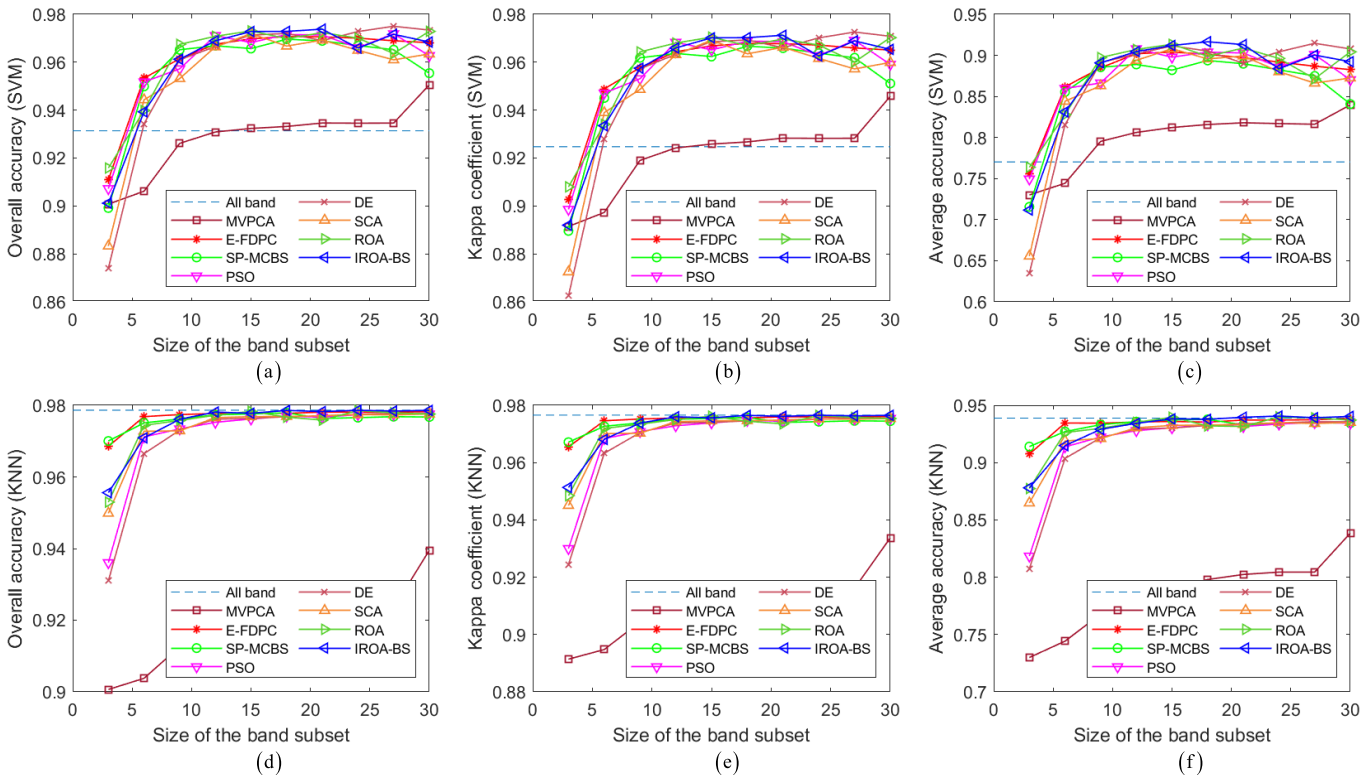



Fig. 7. Classification curves of different number of the selected bands on Pavia Center. (a) OA curves by SVM. (b) Kappa curves by SVM. (c) AA curves by SVM. (d) OA curves by KNN. (e) Kappa curves by KNN. (f) AA curves by KNN.

outperforms others, achieving an OA of 98.88%, Kappa of 98.81%, and AA of 98.73% when applied with the SVM classifier. This exemplifies the IROA-BS's proficiency in band subset selection for classification purposes. Furthermore, the ROA method secures the second position on the SVM classifier with an OA of 98.86%, while the DE method ranks second on the KNN classifier with an OA of 97.50%. This juxtaposition of results from swarm intelligence algorithms against ranking-based methods underscores the former's effectiveness in BS.

*2) Pavia Center:* In this section, a comparative trial conducted at the Pavia Center is elaborated upon. Fig. 7 presents the outcomes spanning from 3 to 30. Utilizing the SVM classifier, an increase in accuracy is observed for all methods when the band subset is sized from 3 to 9. As the OA and Kappa surge beyond 96%, and AA ascends over 89%, the results of the majority of methods reach stability with the selection of 12–21 bands. However, upon exceeding a subset size of 24, the accuracy for most methods exhibits notable instability. In contrast, when using the KNN classifier, there is a rapid accuracy upturn with band subset sizes between 3 and 9, which gradually steadies, achieving an OA approximately at 97% when the band size is beyond 9. It is observed that E-FDPC and SP-MCBS exhibit better performance than IROA-BS when the subset sizes range from 3 to 12. However, the proposed IROA-BS method attains comparable or superior accuracies in larger subset sizes. Conversely, the MVPCA consistently mirrors this trend throughout the trial but maintains its accuracy at a relatively diminished rate.

Table III outlines the comparative analysis of the proposed method against other methodologies across 18 selected band sizes. The table illustrates that the IROA-BS method secures

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TABLE III
CLASSIFICATION ACCURACY OF PROPOSED IROA-BS AND DIFFERENT METHODS ON PAVIA CENTER

| Method | SVM | | | KNN | | |
|---|---|---|---|---|---|---|
| | OA (%) | Kappa (%) | AA (%) | OA (%) | Kappa (%) | AA (%) |
| All band | 93.14±0.08 | 92.47±0.09 | 77.03±0.28 | 97.86±0.05 | 97.66±0.06 | 93.88±0.10 |
| MVPCA | 93.53±0.12 | 92.91±0.13 | 82.03±0.37 | 91.32±0.06 | 90.50±0.06 | 78.29±0.35 |
| E-FDPC | 97.07±0.04 | 96.79±0.16 | 90.12±0.37 | 97.77±0.04 | 97.56±0.04 | 93.53±0.15 |
| SP-MCBS | 96.96±0.10 | 96.67±0.11 | 89.38±0.36 | 97.77±0.16 | 97.55±0.17 | 93.56±0.53 |
| PSO | 97.16±0.24 | 96.88±0.26 | 90.40±0.70 | 97.70±0.12 | 97.47±0.14 | 93.30±0.36 |
| DE | 97.20±0.37 | 96.93±0.40 | 90.53±1.31 | 97.77±0.23 | 97.56±0.25 | 93.37±0.75 |
| SCA | 96.67±0.27 | 96.35±0.29 | 89.57±0.79 | 97.69±0.07 | 97.46±0.08 | 93.27±0.30 |
| ROA | 96.98±0.32 | 96.69±0.35 | 89.89±1.07 | 97.67±0.16 | 97.44±0.17 | 93.20±0.49 |
| IROA-BS | **97.29±0.14** | **97.02±0.15** | **91.65±0.61** | **97.86±0.12** | **97.65±0.13** | **93.78±0.38** |

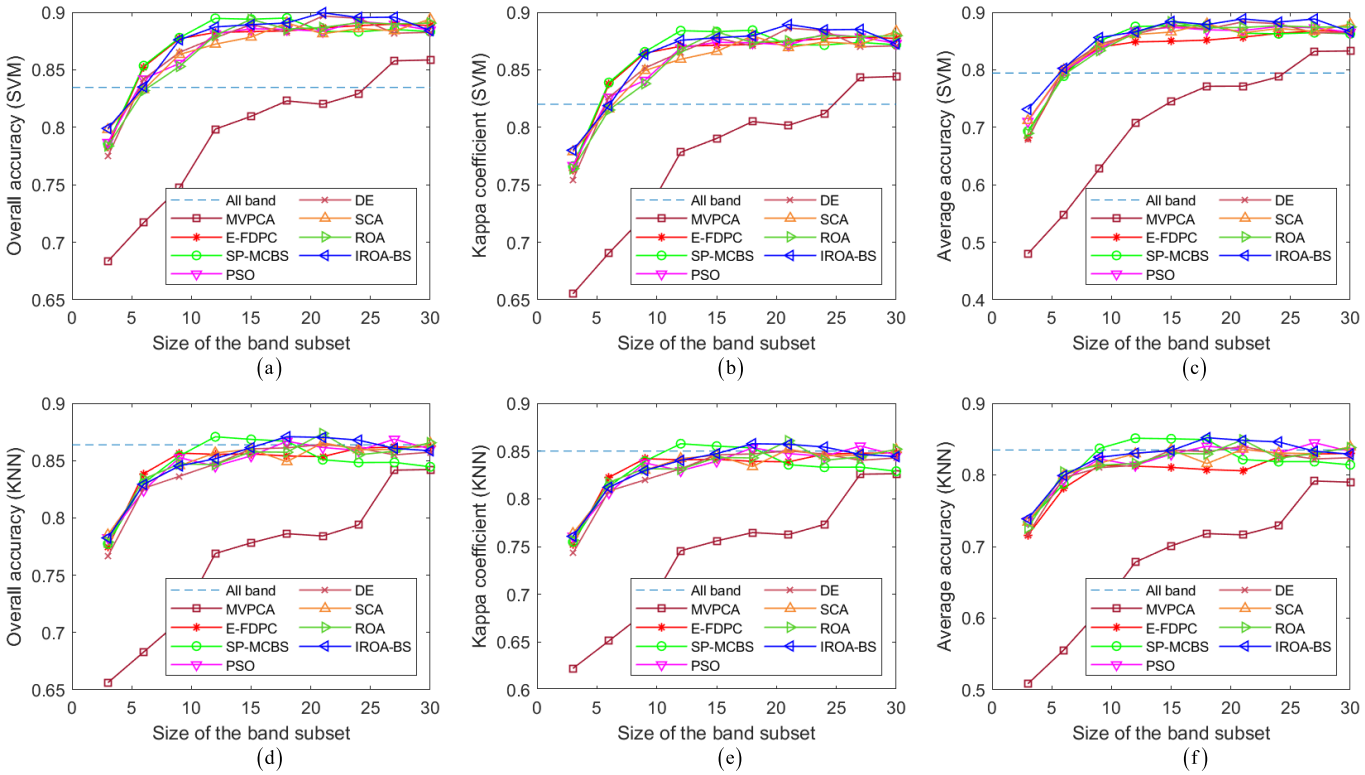



Fig. 8. Classification curves of different number of the selected bands on Pavia University. (a) OA curves by SVM. (b) Kappa curves by SVM. (c) AA curves by SVM. (d) OA curves by KNN. (e) Kappa curves by KNN. (f) AA curves by KNN.

the highest classification metrics with an OA of 97.29%, a Kappa of 97.02%, and an AA of 91.65% when implemented with SVM. Furthermore, when applied with KNN, it achieves an OA of 97.86%, a Kappa of 97.65%, and an AA of 93.78% on KNN. Additionally, the DE method ranks second in performance with an SVM OA of 97.20%. According to KNN results, E-FDPC, SP-MCBS, and DE equally reach an OA of 97.77%, sharing second place jointly. Notably, the standard deviation for E-FDPC is marginally lesser by 0.04% compared to the other investigated methods.

*3) Pavia University:* In this section, we evaluate the performance of various band subsets utilizing the Pavia University dataset. Fig. 8 depicts the classification accuracy curves for different methodologies, with band subset sizes varying from 3 to 30. The depicted accuracy curves indicate that the SP-MCBS's selected bands exhibit the most superior performance when the subset sizes are 9 and 12 for SVM and KNN classifiers correspondingly. Although most methods fall short of SP-MCBS's performance at these specific subset sizes, a notable decline in accuracy for SP-MCBS is observed when the subset size exceeds 18. Conversely, the IROA-BS and DE methods yield outstanding results, outperforming SP-MCBS in OA and Kappa when the band subset size reaches 21 on SVM. Furthermore, the AA of the band subset selected by the proposed IROA-BS method shows superior performance compared to other BS methods when classified by SVM. When analyzed under the KNN classifier, most BS methods have a similar result with three bands. When the sizes ranged

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TABLE IV
CLASSIFICATION ACCURACY OF PROPOSED IROA-BS AND DIFFERENT METHODS ON PAVIA UNIVERSITY

| Method | SVM | | | KNN | | |
|---|---|---|---|---|---|---|
| | OA (%) | Kappa (%) | AA (%) | OA (%) | Kappa (%) | AA (%) |
| All band | 83.46±0.19 | 82.00±0.21 | 79.44±0.41 | 86.38±0.24 | 84.99±0.27 | 83.48±0.35 |
| MVPCA | 82.29±0.31 | 80.50±0.34 | 77.13±0.78 | 78.62±0.33 | 76.47±0.36 | 71.80±0.60 |
| E-FDPC | 88.38±0.42 | 87.20±0.46 | 85.16±0.94 | 85.40±0.54 | 83.92±0.59 | 80.75±1.05 |
| SP-MCBS | **89.50±0.51** | **88.44±0.56** | 87.78±0.56 | 86.69±0.63 | 85.33±0.70 | 84.93±0.58 |
| PSO | 88.52±0.88 | 87.35±0.97 | 86.95±1.22 | 86.15±1.10 | 85.33±1.72 | 84.01±2.09 |
| DE | 88.25±1.74 | 87.05±1.91 | 87.09±1.46 | 86.11±1.34 | 84.70±1.47 | 83.24±1.78 |
| SCA | 88.68±1.04 | 87.54±1.14 | 86.83±1.18 | 84.90±2.28 | 83.37±2.50 | 81.58±3.55 |
| ROA | 88.54±1.44 | 87.37±1.59 | 87.31±1.61 | 85.73±1.47 | 84.28±1.61 | 82.92±2.18 |
| IROA-BS | 89.07±1.36 | 87.95±1.49 | **87.86±1.50** | **87.09±1.82** | **85.76±1.99** | **85.20±3.17** |

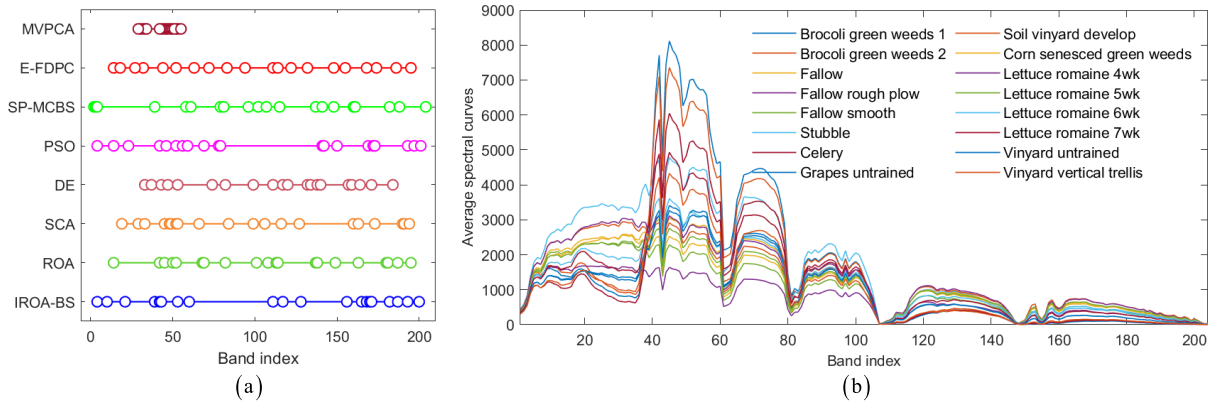



Fig. 9. Distributions of band subset selected by different methods on Salinas. (a) Distributions of band subset. (b) Spectral curves.

from 6 to 15, the E-FDPC and SP-MCBS methods displayed higher classification accuracy than the proposed IROA-BS. However, IROA-BS achieves better results than other comparable methods when the band subset sizes range from 18 to 24.

Table IV delineates the result specifics for the Pavia University dataset employing 18 selected spectral bands. It is evident from Table IV that the SP-MCBS method attains the foremost OA of 89.50% and a Kappa of 88.44% with SVM. The proposed IROA-BS is a close contender, securing the second position with an OA of 89.07% and a Kappa of 87.95%. Significantly, IROA-BS records the best AA at 87.86% on SVM, underscoring the efficacy of the band subsets chosen by the recommended methodologies for the classification of each kind of surface feature. In terms of the KNN classifier, the IROA-BS method outshines with superior results in OA, Kappa, and AA of 87.09%, 85.76%, and 85.20%, respectively. SP-MCBS follows with an OA of 86.69%, Kappa of 85.33%, and AA of 84.93%, claiming the second-best place. The comparative analysis underscores the dominance of the IROA-BS method in classification accuracy over other intelligence methods and ranking-based approaches.

## V. DISCUSSION AND ANALYSES

To extend the analysis of the selected band subset, this section initiates a detailed exploration by comparing the rate of training samples to aspects such as classification accuracy and the duration of the training, and by comparing accuracy of various categories of each HIS dataset, in addition to analyzing the computational time required by the experimental methodologies within the framework of BS.

### A. Subset Component Analysis

To enhance classification proficiency, it is essential to identify critical bands rich in meaningful information, as this underpins one of the benchmarks for assessing the efficacy of BS methods. Figs. 9–11 illustrate the distribution of band subsets across three datasets, selected by the proposed IROA-BS approach and several other methodologies. The subset sizes stand at 20 for the Salinas dataset, 18 for the Pavia Center, and 18 for the Pavia University dataset. As depicted in the figures, MVPCA tends to select a narrow range of bands, often overlooking extensive characteristic spectra within the entire HSI. This leads to the selection of numerous adjacent bands that share excessive similarity and are therefore less effective for feature classification. In contrast, BSs made by methodologies other than MVPCA are more scattered, which promotes more robust classification outcomes.

### B. Category Classification Analysis

In this section, each category classification accuracy of the proposed IROA-BS method and the other BS method are

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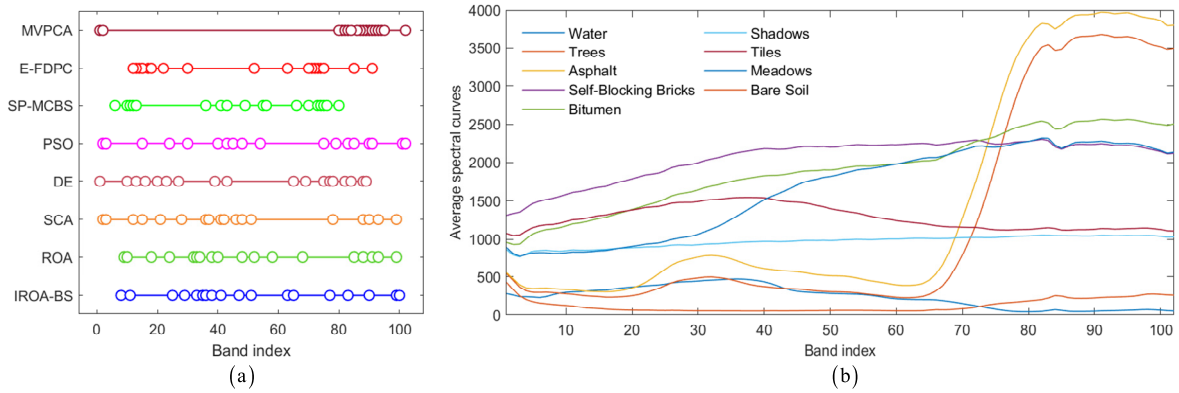



Fig. 10. Distributions of band subset selected by different methods on Pavia Center. (a) Distributions of band subset. (b) Spectral curves.

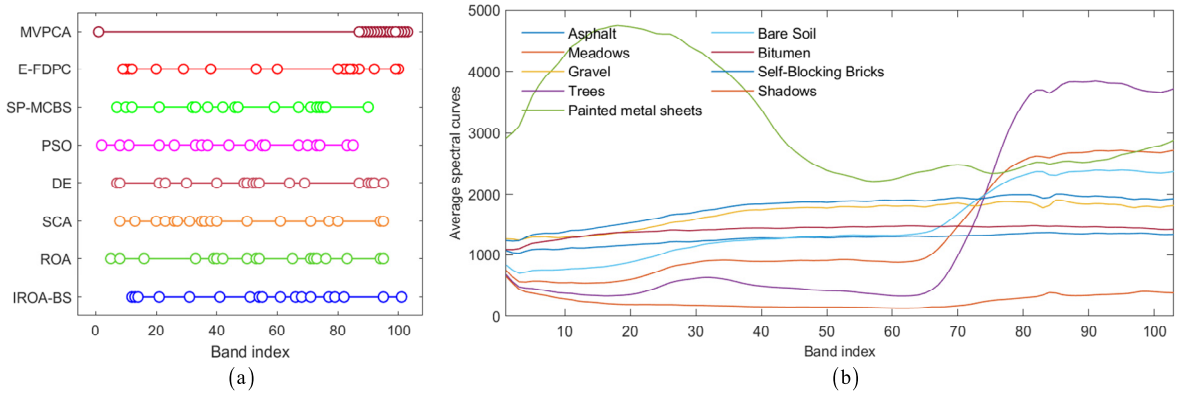



Fig. 11. Distributions of band subset selected by different methods on Pavia University. (a) Distributions of band subset. (b) Spectral curves.

TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON SALINAS BY SVM (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 99.44±0.20 | 98.81±0.29 | 98.24±0.22 | 98.14±0.28 | 99.02±0.94 | 99.20±0.75 | 98.00±1.92 | 98.50±1.28 | **99.45±0.52** |
| 2 | 98.01±0.41 | 98.40±0.52 | 99.36±0.14 | 99.17±0.24 | 99.35±0.14 | **99.47±0.15** | 99.25±0.20 | 99.31±0.15 | 99.42±0.12 |
| 3 | 99.48±0.28 | 99.69±0.08 | 98.41±1.29 | 97.93±1.03 | 99.61±0.41 | 99.56±0.40 | 99.71±0.18 | 99.58±0.22 | **99.78±0.11** |
| 4 | 99.52±0.14 | 99.31±0.28 | 99.18±0.17 | 99.42±0.26 | 99.14±0.61 | 99.58±0.25 | 99.16±0.60 | 99.39±0.49 | **99.76±0.39** |
| 5 | **98.47±0.72** | 97.88±0.34 | 97.89±0.32 | 97.16±0.68 | 97.89±0.43 | 97.80±0.40 | 97.94±0.33 | 97.99±0.33 | 97.77±0.40 |
| 6 | 97.42±0.43 | 99.22±0.29 | 99.42±0.47 | 99.53±0.22 | 99.41±0.35 | 99.53±0.32 | 99.54±0.14 | 99.40±0.20 | **99.66±0.27** |
| 7 | 98.34±0.21 | 98.67±0.20 | 99.32±0.19 | 99.06±0.21 | 99.20±0.20 | 99.34±0.17 | **99.37±0.13** | 99.32±0.18 | 99.24±0.24 |
| 8 | **99.82±0.07** | 98.64±0.19 | 98.51±0.20 | 98.88±0.22 | 98.89±0.29 | 98.98±0.29 | 98.96±0.32 | 99.01±0.19 | 98.96±0.24 |
| 9 | 98.91±0.20 | 99.61±0.09 | 99.47±0.10 | 99.52±0.13 | 99.56±0.21 | **99.67±0.07** | 99.66±0.14 | 99.55±0.15 | 99.61±0.14 |
| 10 | 74.55±1.10 | 91.62±1.03 | 94.76±0.65 | 94.83±0.60 | 95.88±0.83 | 95.67±1.22 | 95.75±1.27 | 95.92±1.37 | **96.19±0.94** |
| 11 | 91.96±2.09 | 94.69±1.33 | 88.94±1.77 | 94.26±1.69 | 97.61±0.72 | 97.21±1.37 | 97.23±1.67 | 97.52±1.10 | **97.89±0.83** |
| 12 | 96.23±1.20 | 99.64±0.28 | 99.43±0.47 | 99.56±0.16 | 99.67±0.14 | 99.56±0.29 | 99.69±0.22 | 99.46±0.38 | **99.71±0.17** |
| 13 | 16.61±2.75 | 95.90±2.53 | 95.44±1.88 | 97.52±0.89 | 96.70±1.32 | 97.32±0.95 | 96.76±1.00 | 96.54±2.60 | **97.62±1.45** |
| 14 | 88.92±1.84 | 95.65±0.91 | 93.81±1.20 | 94.00±1.51 | 95.82±1.73 | 95.79±1.07 | 94.95±1.20 | 95.68±1.43 | **95.87±0.80** |
| 15 | 89.69±1.03 | 98.53±0.22 | 98.96±0.29 | 98.76±0.28 | 98.99±0.40 | 98.98±0.31 | **99.13±0.36** | 98.88±0.29 | 98.92±0.26 |
| 16 | 82.91±1.44 | 93.64±1.21 | 95.36±1.28 | **97.85±0.32** | 96.77±0.58 | 96.22±0.83 | 97.12±0.71 | 96.51±0.89 | 96.92±0.79 |

shown in Tables V–X. As can be seen from Table V, under the SVM classifier, the proposed IROA-BS achieves the highest performance in 9 out of 16 categories. In addition to the classification accuracy of all bands only have the best results in categories 5 and 8. Similarly, refers to the evolution method DE has the highest accuracy in categories 2 and 9, and the swarm intelligence method SCA has the optimal accuracy in categories 7 and 15. Among the 16 categories in Salinas classified by the KNN, the accuracy of each experimental methods witnesses a slight reduction. However, IROA-BS still has the highest accuracy in seven out of 16 categories. While the BS methods of E-FDPC and SP-MCBS achieve their

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TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON SALINAS BY KNN (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 97.62±0.65 | 97.62±0.66 | 96.58±0.76 | 96.70±0.36 | 97.18±1.00 | 97.10±0.73 | 97.02±1.00 | 96.57±0.77 | **98.17±0.81** |
| 2 | 99.00±0.18 | 98.76±0.25 | 99.02±0.17 | 99.29±0.13 | 98.82±0.31 | 98.98±0.38 | 98.85±0.38 | 98.96±0.20 | **99.37±0.30** |
| 3 | 99.78±0.06 | 99.44±0.17 | 99.79±0.04 | 99.57±0.15 | 99.69±0.17 | 99.68±0.28 | 99.79±0.15 | 99.38±0.55 | **99.83±0.13** |
| 4 | 99.31±0.12 | 98.90±0.18 | **99.41±0.16** | 98.75±0.18 | 99.25±0.18 | 99.27±0.12 | 99.39±0.19 | 99.18±0.32 | 99.38±0.23 |
| 5 | 97.17±0.55 | 97.66±0.46 | 98.27±0.52 | 97.06±0.27 | 97.86±0.92 | 96.85±0.93 | 97.08±1.11 | 97.22±1.03 | **98.84±0.72** |
| 6 | 99.44±0.09 | 98.95±0.20 | 99.40±0.11 | 99.54±0.07 | 99.40±0.11 | 99.51±0.15 | 99.41±0.10 | 99.37±0.25 | **99.61±0.20** |
| 7 | 98.93±0.11 | 98.70±0.21 | 98.85±0.13 | 99.00±0.14 | 98.78±0.23 | 98.93±0.25 | 98.94±0.23 | 98.71±0.20 | **99.04±0.21** |
| 8 | **97.17±0.24** | 94.09±0.38 | 96.49±0.31 | 95.82±034 | 96.80±0.73 | 96.90±0.71 | 97.05±0.50 | 96.85±0.60 | 96.53±0.58 |
| 9 | 98.22±0.39 | 96.96±0.47 | 97.83±0.49 | **98.67±0.35** | 98.04±0.39 | 97.81±0.36 | 98.17±0.41 | 97.58±0.39 | 97.73±0.45 |
| 10 | **92.40±1.18** | 84.61±1.16 | 88.04±1.36 | 89.67±0.91 | 91.17±1.19 | 90.76±1.47 | 91.45±1.83 | 90.80±2.39 | 91.65±1.44 |
| 11 | 98.24±0.60 | 97.72±1.24 | 98.00±1.16 | 94.99±1.03 | 98.24±1.11 | 98.22±1.23 | 97.93±1.29 | 98.02±1.03 | **98.46±0.54** |
| 12 | **99.99±0.04** | 99.68±0.42 | **99.99±0.04** | 99.85±0.20 | 99.98±0.05 | 99.88±0.25 | 99.98±0.05 | 99.93±0.12 | 99.98±0.04 |
| 13 | 97.05±0.25 | 97.08±0.36 | 97.18±0.22 | **97.35±0.31** | 97.14±0.23 | 97.17±0.29 | 96.99±0.19 | 96.93±0.13 | 97.14±0.20 |
| 14 | **93.88±0.62** | 92.25±1.28 | 93.01±0.59 | 93.26±0.32 | 93.52±0.73 | 93.41±0.69 | 93.62±0.71 | 93.58±0.90 | 93.58±0.88 |
| 15 | **98.48±0.25** | 96.15±0.68 | 97.85±0.32 | 98.35±0.28 | 98.02±0.71 | 98.14±0.54 | 98.11±0.46 | 97.91±0.65 | 97.99±0.56 |
| 16 | **96.90±0.52** | 91.70±1.03 | 95.17±0.98 | 96.72±0.41 | 95.68±1.51 | 96.09±1.77 | 96.01±1.10 | 96.32±1.47 | 95.96±1.86 |

TABLE VII
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON PAVIA CENTER BY SVM (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 98.94±0.09 | 99.00±0.07 | 99.86±0.04 | 99.89±.0.03 | 99.87±0.05 | 99.89±0.05 | 99.89±0.10 | 99.91±0.04 | **99.92±0.06** |
| 2 | 95.89±0.37 | 94.22±0.83 | 93.31±0.62 | 93.31±1.36 | 94.42±1.09 | 94.59±1.25 | 94.02±0.69 | 94.41±0.76 | **96.34±1.26** |
| 3 | **90.45±1.01** | 67.58±1.94 | 81.17±1.77 | 75.43±5.84 | 83.47±2.29 | 82.87±2.26 | 83.19±1.36 | 83.57±2.53 | 83.82±2.34 |
| 4 | 56.48±1.56 | 66.54±2.03 | 76.23±1.97 | 74.42±2.00 | 74.45±3.46 | 74.71±2.38 | 73.92±2.86 | 75.65±2.48 | **77.69±2.33** |
| 5 | 89.59±0.82 | 74.82±1.20 | **94.55±0.58** | 94.52±0.59 | 94.31±1.10 | 94.43±0.57 | 94.26±0.71 | 94.48±0.95 | 94.44±0.90 |
| 6 | **99.58±0.09** | 78.53±0.58 | 90.98±0.51 | 91.55±0.67 | 91.88±1.07 | 91.61±0.93 | 91.88±0.62 | 91.64±1.30 | 92.14±0.78 |
| 7 | 51.36±1.07 | 70.38±0.78 | 89.16±0.65 | 89.09±1.17 | 88.97±2.02 | 89.38±1.51 | 88.46±2.48 | 89.50±1.63 | **90.37±1.31** |
| 8 | 97.93±0.14 | 97.88±0.15 | 99.40±0.06 | 99.26±0.08 | 99.36±0.09 | 99.40±0.06 | 99.37±0.13 | 99.36±0.08 | **99.40±0.04** |
| 9 | 13.93±0.74 | 77.76±1.37 | **87.08±1.22** | 83.57±2.69 | 83.14±3.74 | 84.90±2.96 | 83.69±4.43 | 85.43±2.97 | 86.87±2.57 |

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON PAVIA CENTER BY KNN (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 99.98±0.01 | 99.02±0.10 | 99.97±0.01 | 99.95±0.01 | 99.96±0.01 | 99.97±0.02 | 99.96±0.02 | 99.96±0.02 | **99.99±0.01** |
| 2 | 93.61±0.46 | 93.12±0.56 | **94.34±0.69** | 90.91±0.44 | 93.77±0.95 | 93.66±0.98 | 93.99±0.74 | 93.77±0.86 | 93.03±0.76 |
| 3 | 86.86±1.00 | 74.93±1.43 | 80.54±1.95 | **90.36±1.10** | 83.45±2.01 | 84.02±3.03 | 83.88±2.55 | 84.24±1.85 | 87.59±1.38 |
| 4 | 83.03±1.41 | 61.54±1.99 | 82.62±1.59 | 80.64±2.95 | 80.74±2.19 | 80.67±1.41 | 81.42±1.84 | 80.76±2.49 | **83.27±2.22** |
| 5 | 94.84±0.34 | 68.32±1.34 | 93.80±0.34 | 92.96±0.71 | 93.58±0.85 | 94.30±0.52 | 94.00±0.75 | 94.01±0.60 | **94.91±0.73** |
| 6 | 96.19±0.34 | 53.41±0.65 | 96.01±0.38 | 95.80±0.34 | 96.16±0.41 | 96.14±0.58 | 96.05±0.61 | 96.11±0.54 | **96.31±0.42** |
| 7 | 91.42±0.58 | 62.58±1.21 | 91.49±0.76 | 90.56±0.52 | 90.54±1.58 | 91.26±0.92 | 91.48±0.57 | 91.42±0.93 | **92.02±0.75** |
| 8 | **99.29±0.04** | 97.84±0.11 | 99.25±0.03 | 99.24±0.08 | 99.24±0.11 | 99.24±0.08 | 99.27±0.09 | 99.26±0.11 | 99.25±0.12 |
| 9 | 99.96±0.01 | 67.83±1.27 | **99.97±0.02** | 99.97±0.02 | 99.97±0.03 | 99.95±0.02 | 99.95±0.02 | 99.96±0.02 | 99.95±0.03 |

advantages in category 4 and category 12, and categories 9 and 13, respectively.

Regarding the nine categories in Pavia Center, the proposed IROA-BS demonstrates its advantage compared to the other methods as shown in Tables VII–VIII. Among the classifiers SVM and KNN, IROA-BS has the highest classification accuracy in five out of nine categories. Besides, E-FDPC shows an advantage in categories 5 and 9 with the SVM classifier, and in categories 2 and 9 with the KNN classifier.

Moreover, based on the nine categories on Pavia University, the performance of the proposed method has slightly declined compared to the above datasets as shown in Tables IX and X.

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TABLE IX
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON PAVIA UNIVERSITY BY SVM (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 92.61±0.54 | 92.39±0.43 | 94.84±0.75 | 96.00±0.83 | 96.34±1.02 | 96.74±0.66 | 96.90±0.69 | 96.69±0.60 | **97.49±0.62** |
| 2 | **99.16±0.08** | 94.13±0.47 | 94.87±0.38 | 95.25±0.34 | 94.82±0.64 | 95.18±0.80 | 94.92±0.84 | 94.96±0.70 | 95.45±0.76 |
| 3 | 68.29±1.27 | 56.50±3.09 | 63.38±2.70 | **73.23±3.75** | 66.80±4.68 | 72.15±2.33 | 69.22±2.86 | 68.88±4.85 | 67.44±6.41 |
| 4 | 84.09±1.15 | 82.50±1.16 | 87.46±1.23 | 85.92±1.93 | 87.09±1.18 | 86.82±1.52 | 86.19±2.68 | 87.54±1.72 | **88.14±2.16** |
| 5 | 95.25±1.01 | 97.53±1.17 | 98.94±0.83 | **99.29±0.34** | 98.85±0.61 | 98.85±0.56 | 98.71±0.69 | 99.07±0.40 | 99.01±0.37 |
| 6 | 17.06±0.77 | 56.63±1.79 | 69.60±1.50 | 70.13±5.64 | 66.05±6.27 | 67.17±6.82 | 68.01±5.93 | 65.52±7.65 | **74.61±5.98** |
| 7 | 89.27±1.50 | 57.89±1.47 | 76.99±2.96 | **92.29±0.71** | 88.32±3.72 | 90.38±2.84 | 91.40±2.11 | 91.14±2.11 | 89.39±3.06 |
| 8 | 76.71±2.12 | 54.23±3.90 | 79.64±2.91 | 82.50±3.31 | 84.45±1.79 | 83.42±1.81 | 83.56±1.71 | **84.76±2.64** | 84.01±2.52 |
| 9 | 92.88±2.19 | 87.28±1.79 | 96.95±0.93 | 98.66±0.61 | 98.59±0.65 | 98.11±1.02 | 98.19±0.96 | 97.99±1.39 | **99.18±0.74** |

TABLE X
CLASSIFICATION ACCURACY OF DIFFERENT CATEGORIES ON PAVIA UNIVERSITY BY KNN (%)

| Category | All band | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 93.09±0.45 | 90.99±0.74 | 92.43±0.55 | 93.09±0.95 | 93.22±1.24 | 93.12±1.63 | 93.10±1.80 | 92.98±0.86 | **94.08±0.92** |
| 2 | 96.29±0.23 | 91.59±0.37 | 94.08±0.39 | 93.30±0.84 | 96.01±0.62 | 95.94±0.63 | 96.07±0.35 | 96.45±0.68 | **96.47±0.50** |
| 3 | 52.08±3.64 | 46.56±2.93 | 50.36±3.12 | 56.68±3.59 | 51.64±4.94 | 49.20±8.43 | 49.80±6.35 | 55.67±5.77 | **60.23±5.65** |
| 4 | 84.06±0.70 | 82.10±1.19 | **87.91±1.18** | 84.86±1.55 | 83.47±2.33 | 82.28±2.83 | 82.92±2.83 | 83.64±1.98 | 83.83±2.55 |
| 5 | **99.63±0.06** | 98.43±0.19 | 99.62±0.06 | 99.62±0.08 | 99.61±0.04 | 99.62±0.08 | 99.62±0.04 | 99.60±0.09 | 99.63±0.06 |
| 6 | 51.20±1.20 | 54.81±1.08 | **61.10±1.97** | 60.52±4.39 | 51.08±6.08 | 47.42±6.87 | 47.75±9.04 | 52.31±5.51 | 55.74±2.77 |
| 7 | 80.46±2.51 | 55.78±3.65 | 62.82±6.59 | 89.65±2.35 | 81.80±7.01 | 81.61±7.41 | 77.69±11.25 | 80.43±8.54 | **90.04±1.17** |
| 8 | 82.47±2.17 | 33.48±1.92 | 79.82±2.45 | 82.96±3.27 | **84.32±1.21** | 83.64±2.03 | 83.01±2.91 | 80.47±3.94 | 84.12±2.39 |
| 9 | 98.97±0.25 | 90.35±3.09 | 98.49±0.34 | 98.62±1.63 | 98.53±1.36 | 97.56±2.24 | 98.15±1.63 | 98.71±0.62 | **99.12±0.34** |

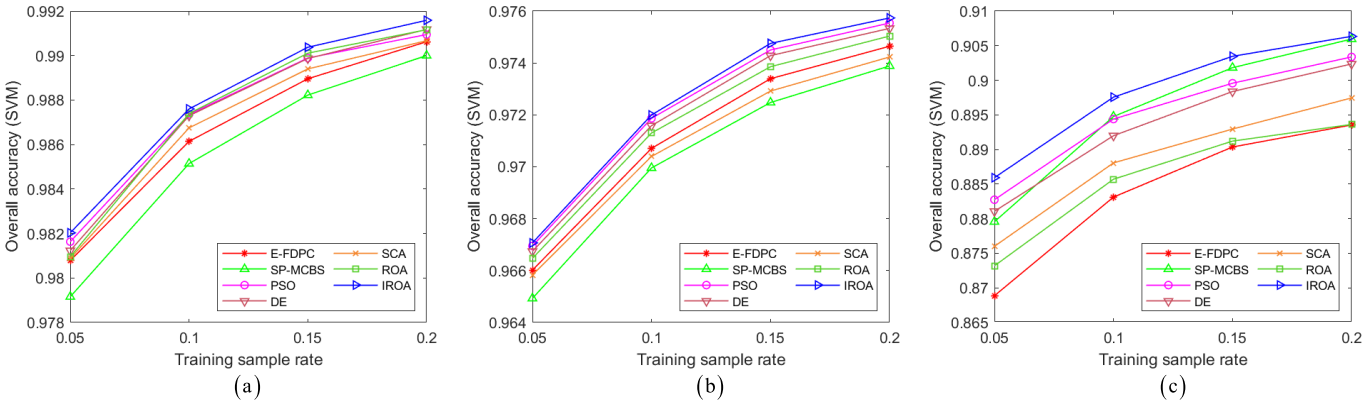



Fig. 12. Classification curves with 20 bands in different training sample rate. (a) OA curves by SVM on Salinas. (b) OA curves by SVM on Pavia Center. (c) OA curves by SVM on Pavia University.

With the SVM classifier, the proposed IROA-BS achieves the best accuracy in four out of nine categories, and SP-MCBS has the best performance in categories 3, 5, and 7. Meanwhile, with the KNN classifier, IROA-BS shows its advantage in five categories. In addition, E-FDPC has the highest accuracy in category 4 and category 6. In summary, the original all band hyperspectral data indeed has the better accuracy in Salinas with the KNN classifier. However, considering the time consumption in the process of the entire band's data, an effective BS method can significantly enhance the classification performance. Based on the table above, the classification accuracy of each category with BS can still achieve or even surpass that of the HSI data with all bands.

### C. Parameter Sensitivity Analysis

Raising the rate of training samples impacts both the classification accuracy and the processing duration of classifiers. Accordingly, this experiment was designed to scrutinize the OA of SVM classifiers, as depicted in Fig. 12. Additionally, Table XI enumerates the training times for a subset of 20 bands chosen through the IROA-BS method on SVM.

As illustrated by the accuracy curves in Fig. 12, variations in the training rate exert minimal influence on the comparative effectiveness of different methods. Notably, as the training rate of the dataset incrementally ascends from 0.05 to 0.2, in steps of 0.05, a corresponding escalation in accuracy is

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TABLE XI
TRAINING TIME OF THE BAND SUBSET SELECTED BY IROA-BS WITH DIFFERENT TRAINING RATE ON SVM (s)

| Dataset | Training sample rate | | | |
|---|---|---|---|---|
| | 0.05 | 0.1 | 0.15 | 0.2 |
| Salinas | 0.8254 | 1.0282 | 1.3072 | 1.6369 |
| Pavia Center | 2.1831 | 6.9173 | 15.2405 | 22.7257 |
| Pavia University | 0.6226 | 0.8677 | 1.3309 | 1.8356 |

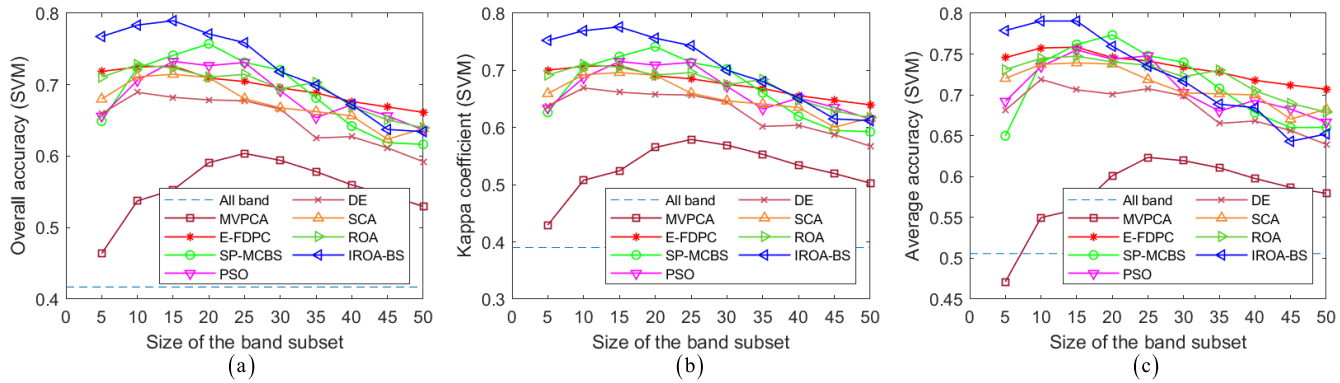



Fig. 13. Classification curves of different number of the selected bands with small sample on Salinas. (a) OA curves by SVM. (b) Kappa curves by SVM. (c) AA curves by SVM.

observed. Specifically, referring to the Salinas results displayed in Fig. 12(a), an increase in training rate from 0.05 to 0.1 correlates with a near 0.6% augmentation in OA. Analogously, the OA results for Pavia Center and Pavia University, as featured in Fig. 12(b) and (c), register enhancements of approximately 0.5% and 1.1%, respectively.

From Table XI, it is evident that as the training rate ascends from 0.05 to 0.2, the training duration for different datasets experiences variegated augmentation. Predominantly, the Pavia Center dataset demands the most extensive training period, attributable primarily to its substantial size. With the training rate advancing from 0.05 to 0.2, the duration is prolonged by almost 20 s. Conversely, the growth in training time imposes a lesser impact on the performance of relatively diminutive hyperspectral datasets, such as those at Salinas and Pavia Universities. Under analogous conditions of training rate, the augmentation in training duration hovers around merely 0.8–1.2 s. Consequently, when reflecting on the efficacy of classification, striking a harmonious equilibrium between training expenditure and accuracy emerges as a quintessential consideration.

In addition, a small sample experiment, with ten training samples of each category, is conducted to analyze the classification efficiency of the selected band subsets. The small sample classification results are presented in Table XII. As can be seen in the table, the training data with all bands exhibit little efficiency in the classification tasks. Meanwhile, the bands selected by MVPCA show some performance improvement compared to the all band condition but acquire the worst accuracy compared to the other BS methods. However, the proposed IROA-BS achieves the best results in the small sample experiment, with an OA of 78.95%, a Kappa of 77.61%, and an AA of 79.05%. Besides, the OA of SP-MCBS is 74.11%, and the Kappa and AA of SP-MCBS are 72.43% and 76.17%, respectively, following the proposed IROA-BS. Fig. 13 shows the classification curves with the size of the band subset ranging from 5 to 50. IROA-BS achieves the highest performance when the subset size is between 5 and 15, and the accuracy decreases faster than other BS methods when the subset size is larger than 15. Nevertheless, the smaller the band subset, the less time-consuming it is, which further demonstrates the BS performance of IROA-BS.

TABLE XII
CLASSIFICATION ACCURACY OF PROPOSED IROA-BS AND DIFFERENT METHODS WITH SMALL SAMPLE ON SALINAS BY SVM

| Method | OA (%) | Kappa (%) | AA (%) |
|---|---|---|---|
| All band | 41.69±3.42 | 39.00±3.44 | 50.57±2.20 |
| MVPCA | 55.29±3.89 | 52.44±4.13 | 55.98±2.85 |
| E-FDPC | 72.61±4.30 | 70.83±4.57 | 75.84±4.23 |
| SP-MCBS | 74.11±3.30 | 72.43±3.52 | 76.17±2.98 |
| PSO | 73.27±6.26 | 71.56±6.66 | 75.50±5.94 |
| DE | 68.23±2.97 | 66.19±3.16 | 70.63±3.75 |
| SCA | 71.46±4.75 | 69.63±5.05 | 73.93±3.77 |
| ROA | 72.35±4.88 | 70.57±5.19 | 74.74±5.52 |
| IROA-BS | **78.95±5.21** | **77.61±5.54** | **79.05±4.51** |

### *D. Computation Time Comparison*

In this section, Table XIII describes the BS duration for all methodologies across three datasets. The comparative analysis underscores that the swarm intelligence approaches register a superior performance, distinguished by reduced computational durations. Specifically, the PSO method marks the shortest duration, with the BS interval proposed by the

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TABLE XIII
COMPUTATION TIME OF DIFFERENT EXPERIMENT METHODS ON EACH HSI DATASETS (s)

| Dataset | MVPCA | E-FDPC | SP-MCBS | PSO | DE | SCA | ROA | IROA-BS |
|---|---|---|---|---|---|---|---|---|
| Salinas | 0.5618 | 0.3325 | 1.5603 | 0.0718 | 0.0986 | 0.0873 | 0.0897 | 0.0934 |
| Pavia Center | 4.0028 | 1.3299 | 8.0150 | 0.0205 | 0.0285 | 0.0240 | 0.0254 | 0.0308 |
| Pavia University | 0.8728 | 0.3441 | 1.9570 | 0.0229 | 0.0299 | 0.0248 | 0.0258 | 0.0292 |

IROA-BS methodology exhibiting a negligible discrepancy of approximately 0.01 s. Moreover, the SP-MCBS methodology necessitates the longest processing durations, incurring additional time exceeding 1.4 s in Salinas and 7.9 s in Pavia Center relative to IROA-BS. Upon a comprehensive assessment, the proposed methodology not only surpasses alternate methodologies in classification efficacy but also leverages computation time efficiency in BS.

## VI. CONCLUSION

In this article, an efficient IROA-BS method is proposed for HSI classification. First, a meticulously designed evaluation function is introduced to improve BS efficacy, thereby procuring band images characterized by pronounced feature separability alongside minimized redundancy. Second, to augment the algorithm's optimization prowess and circumvent the predicament of local optima entrapment, a hybrid evolution module coupled with a multistage mutation module is implemented. Concurrently, the GLS is employed to fine-tune the equilibrium during the optimization phase. The validation of the method's efficacy was undertaken through classification experiments. Furthermore, an analysis encompassing the band subset distribution along with the computation duration for both training and BS phases was conducted. The empirical outcomes underscore the distinguished superiority of the proposed IROA-BS method in addressing the challenges inherent in hyperspectral remote sensing BS. However, IROA-BS still shows unstable performance regarding the stochastic solutions in HSI BS issues.

In the future, the evaluation function of swarm intelligence techniques for the HSI BS issue will be precisely improved to selecting a band subset with lower noise information. In addition, exploring the combination of other BS methods and swarm intelligence will also be significant.

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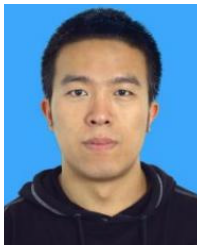
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**Heming Jia** received the B.S. degree in automation and the M.S. degree in control theory and control engineering from Northeast Forestry University, Harbin, China, in 2006 and 2008, respectively, and the Ph.D. degree in system engineering from Harbin Engineering University, China, Harbin, in 2012.

He is currently a Professor with the School of Information Engineering, Sanming University, Sanming, China. His research interests include multiobjective optimization engineering applications and swarm intelligence optimization algorithms.

Dr. Jia is an Editorial Board Member of Biomimetics and Information, respectively. He was also selected as one of the top 2% scientists in the world.



**Zhengbang Li** received the B.S. degree from the School of Computer, Zhuhai College of Science and Technology, Zhuhai, China, in 2022. He is currently pursuing the M.S. degree in computer technology from Fujian University of Technology, Fuzhou, China.

His major research interests include swarm intelligence optimization algorithms, hyperspectral remote sensing image processing, and engineering applications.

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